

LLM, SLM, or Hybrid Utilization: Decision Factors Across Performance, Efficiency, and Domain — A Systematic Literature Review

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Abstract

Large language models (LLMs) deliver strong reasoning and generation performance, but their cost, latency, energy demand, and hardware requirements remain high. Small language models (SLMs) are better suited to edge and privacy-sensitive deployment, yet their capabilities vary across domains. This paper presents a systematic literature review of 54 studies on collaborative SLM–LLM systems spanning hybrid, multi-agent, retrieval-centered, and ensemble architectures, with monolithic LLMs treated as baselines rather than as a primary architecture class. Guided by four research questions on performance, efficiency, orchestration, and domain suitability, we apply a PRISMA 2020-based review design with a three-pass qualitative coding scheme that synthesizes models, architectures, datasets, metrics, latency, cost, energy, and reported limitations. Corpus analysis shows that 70% of studies date to 2025 and hybrid architectures are the most prevalent design; a contribution-type mapping confirms that solution proposals dominate while energy and cost remain markedly underreported relative to accuracy. The literature shows a clear shift toward hybrid and multi-agent designs, but outcome coding reveals that strong results are reliably associated with explicit control layers (routers, retrievers, planners) rather than with model scale. Overall, SLMs are most competitive when paired with routing, retrieval, planning, or fusion mechanisms in narrow, structured, or domain-specific settings.

Keywords

systematic literature review, large language models, small language models, hybrid architectures, multi-agent systems, edge-cloud inference

1. Introduction

Large language models (LLMs) excel at reasoning, coding, and multimodal tasks [1, 2, 3], but their hardware and deployment requirements often confine them to cloud infrastructure or high-end local systems, raising privacy, cloud dependence, and cost concerns [4, 5, 6, 7].

Small language models (SLMs) are attractive for local deployment and lower-cost, data-sovereign operation. Fine-tuning or distillation can also make them competitive on domain-specific tasks [8, 9, 10]. They pair naturally with routing, verification, and planning mechanisms [7, 5, 11]. However, SLMs still trail frontier LLMs on broad reasoning and tasks requiring extensive contextual knowledge [3, 12], which has driven recent work on hybrid and agent-based architectures that combine small and large models for a better capability–efficiency trade-off [13, 14, 15].

This is primarily a systems-design question. Across domains, hybrid and multi-agent architectures aim to improve not only accuracy but also latency, cost, and computational efficiency through edge–cloud verification, expert routing, retrieval, and planner-based orchestration [13, 16, 17, 11, 18, 19, 4]. These developments motivate a closer examination of how collaborative architectures allocate work between small and large models, when monolithic LLM

baselines remain necessary, and when orchestration provides the best balance between capability and efficiency.

Accordingly, this review examines when hybrid or multi-agent SLM–LLM architectures provide a better trade-off between capability and efficiency than monolithic LLM baselines, and under which domain conditions routing, retrieval, planning, and role specialization allow smaller models to contribute effectively.

The main contributions are fourfold: (1) a synthesis of 54 studies on hybrid SLM–LLM architectures, retrieval-centered collaboration, ensembles, and multi-agent workflows; (2) a review structure organized around four practical questions on performance, efficiency, orchestration, and domain suitability; (3) a three-pass qualitative coding scheme that links every synthesis claim to specific primary studies and classifies the provenance of extracted evidence; and (4) a corpus characterization covering publication trends, architecture distribution, domain coverage, and metric reporting rates that supports systematic mapping of the current state of the field.

2. Related Works

Prior secondary studies on small and large language model collaboration can be organized into three groups: a PRISMA-guided systematic literature review, narrative survey papers, and adjacent systematic mapping work. We discuss the contributions of each group on their own terms and then position the present review in terms of scope and methodology.

2.1. Systematic Literature Review

Corradini et al. [20] conduct a PRISMA 2020-guided systematic literature review on the state of SLMs, covering definitions, capability benchmarks, training paradigms (pre-training, instruction tuning, distillation), and early deployment characteristics across diverse architectures and domains. Their study is the only formally protocol-driven predecessor work; it provides a rigorous and reproducible characterization of the SLM landscape as a whole. However, its scope is intentionally broad with respect to the SLM category in general: it does not focus on collaborative SLM–LLM systems as a primary unit of analysis, it does not treat monolithic LLMs as controlled baselines, and it provides limited coverage of the 2025–2026 wave of hybrid and multi-agent orchestration architectures that forms the core of the present corpus. Engineering-level efficiency metrics—latency, cost, and energy—are not extracted as primary outcome variables.

2.2. Narrative Surveys

Three recent surveys address aspects of SLM–LLM collaboration from complementary perspectives [21, 22, 23]. Wang et al. [21] provide a comprehensive taxonomy of SLM architectures and training strategies, including fine-tuning, compression, and distillation, with attention to deployment characteristics and trustworthiness. Chen et al. [22] survey collaboration mechanisms between large and small models, distinguishing distillation pipelines, routing strategies, ensemble methods, and emerging agent-based coordination patterns, and identify key design principles for leveraging complementary model strengths. Wang et al. [23] focus on performance dimensions of SLM–LLM collaboration, reviewing benchmark results across reasoning, coding, and domain-adaptation tasks, and examining cost-effectiveness and cloud–edge privacy trade-offs. These surveys offer valuable breadth and conceptual synthesis that a narrower protocol-driven review cannot replicate. They differ from protocol-based reviews primarily in the traceability of corpus construction: inclusion rationale is not systematically documented through a registered screening procedure, which limits the ability to independently verify corpus completeness—a distinction of transparency rather than of quality.

2.3. Methodological Positioning

A range of methodological frameworks is available for secondary studies in software and systems engineering. Kitchenham and Charters [25] formalize systematic review guidelines with emphasis on structured evidence synthesis and quality appraisal. Petersen et al. [26] introduce the systematic mapping study approach, which aims to characterize a field structurally through publication frequency, research type classification, and topic coverage analysis. Informal narrative synthesis, as employed by the surveys above, serves distinct objectives and produces broader conceptual coverage. PRISMA 2020 [24] was selected for the present study for three reasons specific to the research objectives: (i) the target architecture class—collaborative SLM–LLM systems—requires multi-criterion judgment at screening to separate from adjacent literature, making documented stage-by-stage inclusion decisions essential; (ii) the PRISMA flow diagram and eligibility matrix make corpus construction independently verifiable; and (iii) PRISMA’s structured reporting requirements are well-suited to drawing operational conclusions from heterogeneous empirical evidence. This choice is complementary to, not superior to, the Kitchenham-guided or narrative traditions employed by related works; each tradition serves different secondary-study objectives.

2.4. Scope Differentiation

The present review advances on existing secondary work in four respects. First, **architectural scope**: the review is narrowed exclusively to collaborative SLM–LLM systems (hybrid, multi-agent, retrieval-augmented, and ensemble architectures), with monolithic LLMs treated only as controlled baselines; related surveys treat SLMs more broadly or include studies without a collaborative component. Second, **temporal coverage**: 94% of included studies were published in 2024–2026, capturing recent hybrid edge–cloud and multi-agent orchestration developments that postdate most related surveys. Third, **engineering orientation**: latency, cost, and energy are extracted as primary outcome variables alongside accuracy, making efficiency trade-offs explicit first-class research questions; no related survey reports all four metrics systematically across a documented study corpus. Fourth, **evidence traceability**: a three-pass qualitative coding scheme (Section 3.6) links each synthesis claim to specific primary study identifiers, and a cross-study RQ-to-evidence matrix (Table 2) supports independent verification of conclusions following the concept-centric synthesis approach advocated by Webster and Watson [27].

3. Research Methodology

In this section we will outline the review protocol, including the research questions, search strategy, eligibility criteria, study selection, and data extraction and synthesis process.

3.1. Research Questions

To systematically evaluate the technical and operational trade-offs between collaborative SLM–LLM systems and monolithic LLM baselines, the review protocol is structured around the following four core research questions (RQs):

- **RQ1 (Performance)**: How does the performance (accuracy/efficacy) of multi-agent SLMs or hybrid architectures (LLM + SLM) compare with that of monolithic LLM baselines in specific reasoning and classification tasks?
- **RQ2 (Efficiency)**: What is the impact of hybrid and multi-agent architectures on computational efficiency, specifically regarding inference cost, latency, and energy consumption, compared to monolithic LLM baselines or full-LLM inference?
- **RQ3 (Orchestration)**: What orchestration and routing mechanisms are employed to manage the division of labor between large and small models in collaborative systems?

- **RQ4 (Domain Suitability):** In which specific domains (e.g., healthcare, coding, telecommunications, and edge or multimodal settings) do collaborative SLM-centered architectures achieve performance comparable to monolithic LLM baselines, and where do they fail?

3.2. Information Sources and Search Strategy

To capture a comprehensive and high-quality corpus of contemporary collaborative language model architectures, the literature search was centered on the Web of Science (WoS) Core Collection, with the final search performed on 8 February 2026. Scopus and the ACM Digital Library were also checked with the same core concepts as coverage validation; this check did not identify additional eligible studies that would materially alter the final corpus.

The search strategy was systematically developed by extracting the core concepts from the defined research questions and refining the terminology through initial pilot searches. This iterative process helped identify common synonyms and alternate phrasing used in the literature (e.g., “Compact Model” for SLMs, or “Agentic” for multi-agent systems). To guarantee that every retrieved study examined the intersection of model size, collaborative architectural design, and system evaluation, the final query was structured into four mandatory components and applied to the title, abstract, and keyword fields: (*“Large Language Model*” OR “LLM” OR “Generative AI” OR “Foundation Model*”*) AND (*“Small Language Model*” OR “SLM” OR “Compact Model*” OR “Lightweight Model*”*) AND (*“Multi-agent” OR “Hybrid” OR “Orchestration” OR “Routing” OR “Ensemble” OR “Agentic”*) AND (*“Performance” OR “Accuracy” OR “Efficiency” OR “Latency” OR “Cost” OR “Benchmark”*).

Because the review targets collaborative SLM–LLM systems, all four components were mandatory in the search string. Monolithic LLMs were considered only as baselines reported inside the included studies, not as standalone primary studies.

3.3. Eligibility Criteria

To systematically isolate systems-level engineering research from the broader generative AI literature, retrieved records were evaluated against strict eligibility criteria. A comprehensive breakdown of the inclusion and exclusion parameters applied during the screening phases is presented in Table 1.

Table 1
Inclusion and Exclusion Criteria for Study Selection

Criterion	Include	Exclude
Architecture Scope	Collaborative SLM–LLM systems using routing, retrieval, verification, fallback, or role specialization.	SLM-only papers or monolithic LLM comparisons without a collaborative SLM component; monolithic LLMs were allowed only as baselines inside otherwise eligible studies.
Model Size and Transparency	Identifiable SLM component with reported or inferable size not exceeding 7B parameters; closed-box LLMs allowed only as baseline, teacher, or fallback components.	Primary architecture depends on closed-box/API-bound models with undisclosed details, or no reproducible SLM component can be identified.
Topic Focus	Reports system-level metrics such as performance, latency, memory, cost, or orchestration efficiency.	Focuses only on ethical, sociological, or legal issues without technical benchmarks.
Data Quality	Contains empirical results and experimental validation using benchmarks or custom datasets.	Purely theoretical papers or non-empirical narrative surveys.

3.4. Study Selection Process

The study selection process was guided by the PRISMA 2020 guidelines [24]. The initial search yielded 116 records. During Identification and Screening, the title, abstract, and keywords of all

records were reviewed manually, excluding 36 records that failed the primary eligibility criteria, including non-technical narrative surveys, extended abstracts without full empirical validation, and studies in unrelated domains. The remaining 80 reports advanced to full-text assessment, where 26 additional records were excluded due to model size or architecture mismatch ($n=18$), topic or scope mismatch ($n=5$), or lack of original empirical data or a remaining narrative-survey character ($n=3$). To improve objectivity, inclusion and exclusion decisions at both stages were made by majority vote, with ties resolved through collaborative discussion and escalation to the academic advisors, resulting in a final corpus of 54 primary studies.

3.5. Data Extraction and Synthesis Strategy

To systematically address the research questions, a standardized data extraction protocol was applied to the final corpus of 54 primary studies. For each study, data was extracted into a structured matrix to capture both the functional design of the systems and the empirical evidence used to justify them.

The extracted fields included the specific SLM and LLM models utilized; the overarching architecture type (e.g., Hybrid, Multi-Agent, Retrieval-Augmented Generation (RAG), Ensemble); orchestration mechanisms; evaluation metrics; baseline models; and quantitative performance indicators (accuracy, latency, cost, and energy). To ensure analytical rigor, each extracted variable was explicitly mapped to one or more of the core Research Questions (RQs). This mapping, together with concise field definitions, is detailed in Table 2.

The full comparative analysis table and the comprehensive data extraction spreadsheet are available at <https://bit.ly/SLM-LLM-Data-2026>; for quick reference, the Study ID to citation mapping is as follows: **Primary Studies Mapping:** S1: [1], S2: [2], S3: [3], S4: [4], S5: [5], S6: [6], S7: [7], S8: [8], S9: [9], S10: [10], S11: [11], S12: [12], S13: [13], S14: [14], S15: [15], S16: [16], S17: [17], S18: [18], S19: [19], S20: [28], S21: [29], S22: [30], S23: [31], S24: [32], S25: [33], S26: [34], S27: [35], S28: [36], S29: [37], S30: [38], S31: [39], S32: [40], S33: [41], S34: [42], S35: [43], S36: [44], S37: [45], S38: [46], S39: [47], S40: [48], S41: [49], S42: [50], S43: [51], S44: [52], S45: [53], S46: [54], S47: [55], S48: [56], S49: [57], S50: [58], S51: [59], S52: [60], S53: [61], S54: [62]. Following data extraction, a narrative synthesis approach was utilized. Because the included studies operate across diverse domains (e.g., telecommunications, healthcare, software engineering) and report varying metrics, a purely statistical meta-analysis was not feasible. Instead, studies were grouped by their architectural taxonomy, allowing for a comparative analysis of how different orchestration methods and domain constraints impact the performance and efficiency trade-offs between large and small models.

3.6. Synthesis Protocol and Qualitative Coding

To ensure that synthesis claims are systematically derived from and traceable to primary evidence, data extraction was followed by a three-pass qualitative coding procedure [26]. This procedure standardizes interpretive synthesis decisions across heterogeneous studies and classifies the provenance of extracted statements to distinguish direct evidence from analytical inference.

First pass—Architecture and Orchestration Categorization. Each study was independently coded by two reviewers for (i) primary architecture type: Hybrid, Multi-Agent, Retrieval-Augmented, Ensemble, or Combined; and (ii) dominant orchestration pattern: token-level verification, query-level routing, task decomposition, retrieval augmentation, multimodal fusion, or role specialization. Inter-rater disagreements were resolved through structured discussion, following the same consensus procedure applied during screening (Section 3.4), with escalation to academic advisors for unresolved cases.

Second pass—Domain and Metric Coverage Coding. Each study was classified by primary application domain (Healthcare/Medical, Telecommunications/Networks, Software Engineering, Education/Assessment, Industrial/Edge/IoT, or General NLP) and by metric

Table 2

RQ $\times$ data extraction matrix. Check marks indicate direct support; Arch. = architecture, Orchestr. = orchestration, and Base. = baseline.

Field	Description	RQ1	RQ2	RQ3	RQ4
SLM(s)	Specific small language model(s) evaluated, fine-tuned, or deployed.	✓			✓
LLM(s)	Large language model(s) used as a baseline, teacher, or fallback.	✓			✓
Arch. Type	Overall system design (e.g., hybrid, multi-agent, RAG, ensemble).	✓			✓
Orchestr. Method	Coordination mechanism such as routing, decomposition, fusion, or voting.			✓	✓
Agent Count	Number of distinct agents, experts, or role-specific components.		✓	✓	✓
Task Type	Downstream application, domain, or problem setting evaluated.	✓		✓	✓
Dataset	Benchmark, open-source corpus, or proprietary data used in evaluation.	✓		✓	
Metrics	Evaluation criteria used to assess performance and efficiency.	✓	✓	✓	✓
LLM Base.	Standalone performance score of the comparison large model.	✓		✓	✓
SLM Base.	Standalone performance score of the comparison small model.	✓		✓	✓
Accuracy	Final performance score achieved by the proposed architecture.	✓		✓	
Latency (ms)	Reported inference delay, throughput, or time-to-response.		✓	✓	✓
Cost (\$)	Reported API cost or computational savings versus full LLM use.		✓	✓	✓
Energy	Power use, joules-per-token, or carbon-related efficiency metric.		✓	✓	✓
Findings	Main study takeaway on SLM–LLM collaboration or orchestration.	✓	✓	✓	✓
Limits	Reported constraints, failure modes, or deployment bottlenecks.		✓	✓	

availability: (a) performance-only; (b) performance plus at least one efficiency metric; or (c) full-metric reporting including accuracy, latency, cost, and energy. This coding produced the distribution statistics reported in Section 4.1.

Third pass—Performance Outcome and Evidence Provenance Classification. Each study’s primary empirical finding was classified as one of: (a) outperforms monolithic SLM baseline; (b) matches monolithic LLM baseline; (c) outperforms LLM baseline; or (d) mixed or task-dependent results. Concurrently, each extracted data item was assigned a provenance label:

- *Direct numerical extraction*: a quantitative result reproduced verbatim from the primary paper;
- *Direct textual extraction*: a qualitative claim or limitation reproduced verbatim or with minor paraphrase, with page reference;
- *Analytical synthesis*: a pattern, limitation, or implication derived by the review team from architectural descriptions or indirect evidence in the source paper, where the primary study does not make the statement explicitly.

In the supplementary extraction spreadsheet (<https://bit.ly/SLM-LLM-Data-2026>), cells derived by analytical synthesis are annotated “[Synthesized]” to distinguish them from directly extracted claims. This distinction is particularly relevant for limitation statements: where an author

explicitly states that a metric was not measured, the limitation is recorded as a direct textual extraction; where a limitation is inferred by the reviewers from architectural properties not discussed by the authors, it is recorded as an analytical synthesis. For example, in S1 the latency field records “not reported numerically” as a direct textual extraction, while the associated limitation note regarding production-time overhead from iterative refinement loops is recorded as an analytical synthesis based on the described pipeline.

Following Webster and Watson [27], synthesis conclusions in Section 4 are organized concept-centrally rather than study-by-study: each concept (orchestration pattern, efficiency trade-off, domain constraint) is linked to the studies providing evidence for it via the S1–S54 identifier scheme. Design pattern claims in Section 4.7 were retained as primary implications only when supported by at least four independent primary studies reporting consistent positive outcomes under comparable operational constraints; the supporting study set for each pattern is stated explicitly in that section.

4. Findings

4.1. Study Characteristics and Corpus Overview

Before addressing the research questions, a characterization of the 54-study corpus provides context for interpreting the synthesis.

Temporal distribution. The corpus is concentrated in recent years: one study dates to 2023 (2%), 13 to 2024 (24%), 38 to 2025 (70%), and two to early 2026 (4%). The dominance of 2025 publications indicates an emergent and fast-moving research area in which collaborative SLM–LLM architectures are still consolidating. Publications cluster around two conference cycles (ACL/EMNLP and AAAI/WWW) and several open-access engineering journals, with a notable share appearing first as arXiv preprints later published in conference or journal proceedings.

Architecture distribution. Hybrid architectures—comprising token-level verification, query-level routing, and task-decomposition systems—constitute the largest group. Multi-agent systems form the second-largest group, followed by retrieval-augmented designs and ensemble approaches. Several studies combine multiple patterns (e.g., multi-agent systems that also employ RAG), so per-study architecture labels are not mutually exclusive; the full distribution table is available in the supplementary spreadsheet. Hybrid designs grew as a proportion of the corpus between 2024 and 2025, while ensemble-only studies remained stable. Multi-agent architectures increased substantially in 2025, consistent with the broader uptake of agentic frameworks in the literature.

Domain distribution. Healthcare and clinical applications account for the largest single domain (approximately 14 studies), spanning radiology labeling, clinical summarization, TCM classification, and biomedical agents. Software engineering and code-related tasks form the second-largest group (approximately 8 studies), covering code correction, static analysis, comment generation, and robot introspection. Telecommunications and network optimization (approximately 6 studies), edge/IoT/industrial settings (approximately 6 studies), and education/assessment tasks (approximately 4 studies) follow. The remaining studies address general NLP tasks or cross-domain benchmarks. Domain distribution is uneven; healthcare and NLP are substantially better represented than education, transportation, and emerging multimodal settings.

Metric reporting. The second coding pass (Section 3.6) confirmed that performance metrics (accuracy, F1, task-specific scores) are reported in nearly all 54 studies. Latency is reported in approximately one-third of the corpus, cost in fewer than one-quarter, and direct energy measurement in only a handful of studies. This uneven reporting profile reflects the field’s current maturity and directly constrains what cross-study efficiency comparisons are possible; this limitation is discussed in full under RQ2 (Section 4.3).

Contribution types. Classified following Petersen et al. [26], the corpus is dominated by *solution proposals* (novel systems presented with experimental evaluation), with a smaller set of

evaluation studies (comparative analyses of existing models or frameworks) and a few *validation studies* (proof-of-concept designs without full benchmark evaluation). The predominance of solution proposals is consistent with the field’s early stage: few standardized benchmarks exist for collaborative SLM–LLM systems, and most papers define their own evaluation protocols.

4.2. RQ1: Comparative Performance of Hybrid and Multi-Agent Systems

The review shows that both hybrid and multi-agent architectures can outperform standalone small-model baselines, but their advantages emerge through different forms of coordination rather than parameter scaling alone. Hybrid systems achieve their strongest gains when computation is selectively distributed between local small models and stronger remote models. Token-level edge–cloud collaboration improves small-model performance without requiring full-query escalation: draft-verification, low-confidence routing, and uncertainty-aware skipping allow small models to approach LLM-level quality while invoking the larger model selectively (S13, S48, S49). At the sub-task level, structured task decomposition and cloud–edge allocation also maintain reasoning accuracy comparable to the best baselines (S14). In domain-specific settings, hybrid orchestration can outperform larger general-purpose models: a fine-tuned Phi-2 (2.7B) combined with retrieval, reranking, hybrid search, and long-context support achieves 84.20% accuracy, exceeding GPT-4o mini (82.05%) and GPT-4o (80.80%) on telecom question answering (S17).

Multi-agent systems, by contrast, show their comparative strength when performance depends on role specialization, critique, verification, and distributed reasoning. A three-agent Phi-3-based architecture performs on par with human experts in conceptual genomics tasks, although its gains weaken on more complex code-generation workflows, indicating that multi-agent benefits remain task-sensitive (S52). In reasoning-intensive question answering, a Proponent–Opponent–Arbitrator structure built on compact models such as Qwen2.5 0.5B and DeepSeek-R1 1.5B surpasses both single-agent reasoning and vanilla LLM baselines (S43). Likewise, assistant–checker collaboration supported by memory and feedback loops enables small-model multi-agent systems to approach or exceed larger baselines across database, knowledge-graph, and code-correction tasks (S29).

Summary (RQ1). Hybrid and multi-agent architectures consistently outperform monolithic SLM baselines and can match or exceed LLM baselines in narrower settings. The performance advantage is driven by orchestration strategy—selective escalation, role decomposition, verification loops—rather than by model size alone. Gains are strongest in structured, domain-specific tasks and weakest in open-ended reasoning that requires broad contextual knowledge.

4.3. RQ2: Efficiency, Cost, Latency, and Energy Trade-offs

Efficiency motivates many hybrid and multi-agent systems, but the impact on cost, latency, and energy depends heavily on orchestration. Hybridization is not a universal efficiency solution; it often shifts bottlenecks rather than eliminating them.

Regarding inference cost and token consumption, architectures employing selective escalation or task decomposition drastically reduce reliance on massive standalone LLMs. S49 achieves an over 80% cost reduction compared to full LLM inference by routing fewer than 7% of generated tokens to the cloud. Similarly, S13 uses a token-level draft-verification method to match the quality of a 70B parameter LLM while consuming only 25.8% to 31.2% of standard cloud inference costs. Query-level decomposition yields similar savings: S14 cuts average API costs by 83.57% via parallel reasoning. Computational savings are also heavily emphasized through token reduction. S40 achieves a 95% reduction in LLM token transmissions via federated uncertainty-aware routing, and S15 eliminates redundant multi-round communications through an SLM-generated directed acyclic graph, reducing token consumption by 35% to 70%. In dialogue state tracking, S16 highlights that a retrieval-based router reduces overall computational FLOPs by approximately 62%.

Conversely, the impact on latency is highly polarized. In optimal scenarios, relying on lightweight models accelerates throughput: a fine-tuned SLM in a hybrid classification pipeline (S10) operates at 7.41 ms on a GPU, and uncertainty-aware skipping in an edge–cloud setup provides 2.54× faster token throughput (S48). However, orchestration mechanisms frequently introduce new bottlenecks. Planners and retrievers add temporal overhead, as shown by the adaptive retrieval planner in S11 (1.42 seconds versus 0.95 seconds for naive retrieval) and by cross-encoder reranking in S17. Multi-agent collaboration also multiplies inference cycles: S43 reports a 1.8× inference-time overhead for adversarial debate, while complex agentic workflows can push average response latency to 107 seconds (S31). Processing time is further affected by hardware and modality constraints, including constrained local hardware, dual-path video sampling, CPU-only ensembles, and key-value cache recomputation at the edge (S20, S23, S50, S5).

Energy consumption remains the least documented metric in the reviewed studies, and most reported efficiency gains are resource proxies rather than direct energy measurements. The clearest energy-centered evidence comes from S4, which argues for energy-per-token as a more informative metric and shows that forcing a small model to execute complex reasoning workflows, such as Chain-of-Thought, can substantially increase its per-query energy use. In some settings, a larger model prompted more directly is therefore more energy-efficient than a smaller model that generates a long reasoning trace. By contrast, S29 reports a 26.4% reduction in computational redundancy through dual-scale memory, which may indicate lower resource demand but does not by itself constitute a direct energy benchmark. Taken together, the review suggests that energy efficiency depends on orchestration and generation length as much as on model size, and that stronger claims require standardized, hardware-specific measurement protocols.

Summary (RQ2). The strongest efficiency gains—often exceeding 80% cost reduction—come from token-level routing and selective escalation. Latency effects are polarized: local SLM inference is fast, but orchestration layers (retrievers, planners, multi-agent cycles) frequently introduce overhead that negates throughput gains. Energy is the least reported metric, and available evidence shows that generation length and reasoning depth matter as much as model size for per-query energy use.

4.4. RQ3: Orchestration and Routing Mechanisms

The studies contain a broad set of orchestration mechanisms, but they can be organized into a small number of recurring families. Across the included studies, performance differences are explained less by absolute model size than by how effectively the system retrieves evidence, routes uncertainty, and divides labor across components.

Token-level orchestration is most visible in verification-oriented hybrid systems. S13 uses the SLM as a drafter and the LLM as a verifier on difficult tokens. S48 extends this pattern with uncertainty-aware skipping so that likely acceptable SLM tokens can bypass the remote model. S49 reaches a related design point by routing only low-confidence tokens for external inference. Taken together, these studies show that token-level collaboration provides fine-grained control over answer quality, but it can also introduce communication and synchronization overhead when escalation rates increase.

Query-level routing appears when the system selects a processing path for the full input rather than for individual tokens. S16 exemplifies retrieval-based expert routing at the dialogue level, while S10 shows that instance-level model selection can also be effective in customer-review analysis. These designs are most useful when “easy” and “hard” cases can be separated in a semantically meaningful way before full generation begins.

RAG is a major control mechanism across the included studies even when the architecture is classified as hybrid rather than purely RAG. S17, S11, S39, and S47 use retrieval not only to expand context but also to constrain the action space of a smaller model. In these studies, retrieval acts as a coordination layer that narrows ambiguity, grounds the answer in domain

evidence, and reduces reliance on raw model scale alone.

Task decomposition is increasingly important in the recent literature. S14 uses a task decomposer, dependency graph, and plug-and-play adapter to allocate subtasks between local and remote models. S53 uses a feedback-integrated workflow that gathers structured feedback from multiple perspectives before revising the prompt of the smaller model. Related role-based decomposition also appears in S23, S29, and S52. Across these studies, the common idea is that small models remain viable when the workflow can be partitioned into specialist stages rather than handled as a single undifferentiated generation step.

Orchestration frequently extends beyond unimodal text processing to integrate diverse data modalities and specialized neural architectures. Several frameworks are designed to fuse language models with non-text representations and structured data. For example, S20 combines coarse-grained and fine-grained visual semantics through prompt-guided video sampling, while S8 fuses language-model embeddings with graph-convolutional features for molecular-property prediction. This heterogeneous routing is particularly prevalent in applied domains such as medicine; S19 orchestrates convolutional networks (ConvNeXt) for anatomical localization alongside a self-reflective LLM agent, and S34 employs a late-fusion approach to combine structured gradient boosting models (EHR tabular data) with unstructured clinical text analysis. In these settings, orchestration operates as a structured fusion mechanism, ensuring that visual, spatial, or mathematical data is processed by the optimal specialized architecture before language generation occurs.

Summary (RQ3). Orchestration mechanisms cluster into five recurring families: token-level verification, query-level routing, RAG, task decomposition, and multimodal fusion. Each family addresses a distinct allocation problem—confidence management, difficulty separation, context grounding, subtask specialization, and modality integration—and the choice among them is largely determined by the deployment constraints and data modalities of the target domain.

4.5. RQ4: Domain Suitability and Application-Specific Constraints

The choice between hybrid and multi-agent systems is rarely domain-agnostic. Privacy mandates, latency budgets, and verification demands largely determine the orchestration method. Across the 54 primary studies, distinct architectural preferences emerge across four core applied domains.

In fields with strict privacy constraints or high-stakes accuracy requirements, hybridization is driven by the need to keep data local while grounding the SLM in verifiable facts. Local SLMs fine-tuned on clinical corpora and augmented by RAG are therefore well suited to medical reporting and clinical classification tasks (S19, S30), while few-shot open-source models can be competitive for structured radiology labeling (S12). Related healthcare studies also show that local ensembles, tabular transformations, and distillation can support diagnostic or summarization workflows without relying on a fully remote LLM (S28, S33, S41). In scientific settings, structurally augmented SLMs can be effective when language representations are fused with domain-specific models, as in molecular property prediction with graph-convolutional features (S8).

Tasks requiring rigid syntax, iterative refinement, and logical debugging benefit from multi-agent or feedback-driven SLMs rather than zero-shot massive LLMs. Pedagogy, assessment, code correction, and database tasks favor workflows in which small models receive structure, memory, or explicit feedback (S6, S29, S32). In software and bioinformatics settings, role-specific agents can mimic developer-reviewer workflows and generate or revise code offline, although broad static analysis still favors larger models when the task requires open-ended type inference (S24, S2, S52, S53).

Telecommunications architectures prioritize latency budgets, bandwidth allocation, and domain-specific vernacular, making hybrid edge–cloud systems and retrieval-grounded SLMs the dominant pattern. Telecom QA benefits from fine-tuning, retrieval, reranking, and long-context support because these mechanisms ground smaller generators in technical standards and network terminology (S17, S47). Beyond QA, latency optimization and bandwidth reduction

drive hybridization through lightweight prediction, uncertainty-aware skipping, and sub-task decomposition (S37, S48, S14).

In edge, cyber-physical, and multimodal settings, orchestration is governed by hardware limits and input modality. Mission-critical aerospace and geospatial workflows rely on local or distributed agents to process operational data under deployment constraints (S23, S45). Industrial video monitoring and cross-modal sensing benefit from heterogeneous pipelines in which local models handle coarse or sensor-side processing while larger multimodal models are invoked only for complex cases (S20, S1). Overall, these domains show that collaborative SLM–LLM systems are strongest when the workflow can be partitioned according to privacy, latency, modality, or verification needs.

Summary (RQ4). Domain-specific constraints—privacy mandates in healthcare, latency budgets in telecom, hardware limits in edge deployments, and syntax rigidity in code tasks—are the primary drivers of architecture selection. Collaborative SLM–LLM systems deliver their strongest results when the workflow can be partitioned to align each component with the constraint it best addresses.

4.6. Cross-RQ Synthesis

Viewed together, the four research questions suggest a consistent systems principle: hybrid language-model design works best when the large model is treated as a selectively invoked capability rather than the default engine for every token, query, or subtask.

This conclusion is supported by the outcome coding performed in the third pass (Section 3.6): cross-tabulating performance outcome codes with orchestration pattern codes reveals that studies classified as achieving LLM-competitive or superior results (approximately 39 of the 54 studies) consistently feature at least one explicit control layer. Conversely, studies showing mixed or negative results most frequently involve orchestration overheads—additional latency from multi-round agent cycles or retrieval pipelines—that are not offset by performance gains. Strong results are therefore reliably associated with explicit control layers such as routers, planners, retrievers, or fusion modules; the absence of such a layer is the most consistent differentiator between high- and low-performing collaborative designs.

Future progress is therefore unlikely to come only from making SLMs slightly larger or LLMs slightly cheaper; it will also depend on better confidence estimation, task decomposition, retrieval selection, and cross-model scheduling. Methodologically, because accuracy dominates reporting, many studies overstate practical efficiency. The review includes cases where orchestration clearly reduces cost or latency, but also cases where coordination merely shifts the bottleneck. A mature evaluation protocol should therefore report resource metrics as first-class outcomes. Application-wise, the strongest SLM-centered deployments are domain-adapted systems in which the small model is grounded, specialized, or protected by escalation logic. This is why telecom question answering, selected healthcare tasks, optimisation, and industrial monitoring are prominent in the positive findings.

4.7. Implications for Practitioners

For software engineers and system architects, the main design trade-offs are operational rather than purely model-centric. In practice, latency, cost, privacy, and hardware constraints often dominate deployment decisions. Four practical design patterns were identified through frequency analysis of the architecture and orchestration coding results (Section 3.6); each pattern is supported by at least four primary studies reporting consistent positive outcomes under comparable operational constraints, and the supporting study sets are listed below.

1. **Local SLM + RAG** (privacy-constrained settings): When strict data privacy is the main constraint, local SLM + RAG architectures are most suitable for healthcare, telecom, and

defense settings (S17, S23, S19, S30), but their performance depends heavily on retriever quality.

2. **Hybrid edge–cloud with routing** (high-throughput settings): When high throughput is required, hybrid edge–cloud architectures are appropriate for IoT and network traffic scenarios (S37, S40, S48, S49), reducing API costs while requiring careful threshold tuning.
3. **Multi-agent SLM with role decomposition** (logical reasoning tasks): For logical reasoning tasks such as code and mathematics, multi-agent SLM designs can improve accuracy (S29, S15, S43, S52), but they introduce significant latency overhead.
4. **Heterogeneous fusion** (multimodal data): For multimodal data, heterogeneous fusion is better suited to medical imaging and robotics (S8, S20, S19, S1), although the slowest model in the pipeline often becomes the bottleneck.

Three practitioner-facing principles follow: (1) **Design for routing**: orchestration, not model size, drives performance; (2) **Balance latency-cost trade-offs**: real-time tasks favor token skipping, while batch tasks tolerate multi-agent loops; and (3) **Treat RAG as control**: retrieval constrains the SLM action space and grounds outputs in verifiable facts.

5. Threats to Validity

5.1. Internal Validity

Although the review follows a PRISMA-guided workflow, study identification, screening, and extraction still involve researcher judgment, especially for borderline papers in which the SLM component or empirical contribution was not equally central. To reduce this risk, borderline screening decisions were discussed through team consensus, and the synthesis remained tied to the predefined research questions and the same extraction matrix throughout.

5.2. Construct Validity

Core constructs such as “SLM”, “hybrid architecture”, “multi-agent system”, and “efficiency” are not standardized across the literature. A single hybrid label may refer to token-level verification, retrieval-grounded generation, planner-based decomposition, or multimodal fusion, while efficiency may mean latency, FLOPs, API cost, bandwidth reduction, or energy use. This limits direct comparability because studies grouped under the same high-level category may differ substantially in mechanism. To partially limit this, the review separates broad architecture labels from extracted orchestration mechanisms and discusses recurring mechanism families explicitly.

5.3. Conclusion Validity

Cross-study inference is limited by the uneven and inconsistent reporting of evaluation outcomes. Accuracy is reported in most papers, but latency, cost, and energy appear only in a minority, often with different units, hardware assumptions, or deployment settings, which prevents formal meta-analysis and limits claims about efficiency gains across the included studies. Publication and reporting bias may further favor positive hybrid results, so the extraction also recorded reported limitations, and the synthesis emphasizes uneven reporting and task-sensitive trade-offs rather than universal gains.

5.4. External Validity

The study set is based on a Web of Science-centered search and a bounded review window, which improves transparency but may miss relevant studies from other databases, preprint platforms, or fast-moving venues. Although WoS covers many major peer-reviewed journals and conference

proceedings, the field evolves quickly and the current corpus is heavily concentrated in 2025 and in a limited set of application domains. Therefore, the conclusions should be generalized cautiously beyond the covered time period, tasks, and deployment settings.

6. Conclusion

This paper presented a systematic literature review of 54 studies on collaborative SLM–LLM systems across hybrid, multi-agent, ensemble, and retrieval-centered architectures, with monolithic LLMs treated as baselines rather than as a primary study class. Across the four review questions, a consistent pattern emerges: SLMs are most competitive when they are supported by retrieval, routing, uncertainty estimation, task decomposition, prompt optimisation, or multimodal fusion rather than used as isolated replacements for larger models. The literature is stronger in architectural innovation than in evaluation discipline. Many studies report accuracy improvements, but far fewer report latency, cost, or energy in a form that supports fair comparison. The current evidence is therefore sufficient to show that hybridization can work, but often insufficient to show when it is the better deployment choice under realistic resource constraints. The strongest positive results remain concentrated in narrower or more structured settings such as telecom question answering, selected healthcare tasks, optimisation, and industrial monitoring. The energy evidence is especially limited: direct energy measurement is rare, and the clearest energy-centered result shows that when small models are pushed into long Chain-of-Thought traces, their per-query energy use can exceed that of a larger model prompted more directly [4]. Other studies report only proxy improvements, such as reduced computational redundancy [37], which do not establish verified energy savings. Future work should therefore treat energy as a first-class evaluation outcome and report energy-per-token, reasoning length, hardware configuration, and escalation behavior under comparable conditions.

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Declaration on Generative AI. The authors have not employed any generative AI tools in the preparation of this work.

A. Online Appendix

The online appendix contains the full comparative analysis table for the 54 primary studies and the comprehensive data extraction spreadsheet used in this review.

Access link: <https://bit.ly/SLM-LLM-Data-2026>

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